

H-01 Vision

Field	Instructions / Suggested Length	Content (Dr. Chaw Jun Kit)
Lab name in Chinese	Enter the confirmed official full name.	Harmonizing Intelligence Lab (HI Lab)
Lab name in English	Enter the confirmed official English name.	Harmonizing Intelligence Lab (HI Lab)
Vision title	Summarize the long-term direction in 8–20 characters.	Bridging Human Perception and Machine Intelligence at the Edge.
Vision statement	Describe the core focus in 60–120 characters.	We are dedicated to transforming complex data into actionable intelligence through the fusion of Computer Vision, Multi-Agent Systems, and hardware-efficient Edge AI. Our goal is to address real-world challenges, providing highly adaptable AI solutions for digital health, industrial automation, and smart public services.
Additional note (example)	Describe the primary visual.	Nature-inspired design: Visually interweaving organic leaf structures with circuit board patterns, symbolizing the fusion of biological perception and synthetic intelligence, connecting natural growth with edge computing technology.

H-02 Research Direction

ID	Related Content Name	Fixed detail-entry rule
H-02-01	R-02 HEALTH	Always links to the research-direction page (Digital Health).
H-02-02	R-03 EDGE-AI	Always links to the research-direction page (Edge AI).

H-02-03	R-04 AGENT	Always links to the research-direction page (Agent Systems).
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H-03 Technology Advantages

ID	Fixed Mapping	Card rule
H-03-01	A-01 Predictive Analytics & Knowledge Distillation	Display only "Name + Advantage"; fixed link to the Technology Advantages page.
H-03-02	A-02 Large Language Models & Multi-Agent Orchestration	Display only "Name + Advantage"; fixed link to the Technology Advantages page.
H-03-03	A-03 Edge Vision & Lightweight Deployment	Display only "Name + Advantage"; fixed link to the Technology Advantages page.

H-04 Projects

ID	Instructions	Project ID
H-04-01	Enter one project ID.	P-07-06
H-04-02	Enter one project ID.	P-07-08
H-04-03	Enter one project ID.	P-07-05

H-05 Partners

ID	Fixed Mapping	Display rule
H-05-01	N-02-01	Always reads the organization name and logo (Advantech).
H-05-02	N-02-02	Always reads the organization name and logo (HILTI).
H-05-03	N-02-03	Always reads the organization name and logo (Tokio Marine/Dynafront).

H-05-04	N-02-04	Always reads the organization name and logo (Xiamen University Malaysia).
H-05-05	N-02-05	Always reads the organization name and logo (Leeds Beckett University).
H-05-06	N-02-06	Always reads the organization name and logo (Three-Opp).

H-06 Team

ID	Fixed Mapping	Display rule
H-06-01	T-02 PI	Always links to the team page.
H-06-02	T-04-01	Always links to the team page.
H-06-03	T-04-02	Always links to the team page.
H-06-04	T-04-03	Always links to the team page.

H-07 Contact

ID	Related Content Name	Fixed detail-entry rule
H-07-01	Collaboration format ranked first on the home page	Always links to the contact page (Industrial Tech Transfer).
H-07-02	Collaboration format ranked second on the home page	Always links to the contact page (Joint Academic Research).
H-07-03	Collaboration format ranked third on the home page	Always links to the contact page (Talent Recruitment).

R-01 Research Intro

Entry	Instructions / Suggested Length	Content
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Page Title	Enter the official title of the research-direction page.	Core Research Pillars
Homepage Summary	Read-only summary cited by H-02.	Operating at the intersection of Medical AI, On-Device Vision, and Autonomous Orchestration, delivering high-impact solutions for healthcare, smart cities, and industry.
Inner-Page Introduction	Summarize what the lab researches.	HI Lab focuses on translating cutting-edge machine learning algorithms into enterprise-grade applications. Our research is structured into Digital Health (HEALTH), Edge Intelligence (EDGE-AI), and Multi-Agent Systems (AGENT). Through these pillars, we bridge the gap between low-level data perception (edge vision), domain-specific knowledge parsing (medical/industrial modeling), and top-level autonomous decision-making (LLM orchestration).

R-02 HEALTH (Digital Health)

Entry	Instructions / Suggested Length	Content
Name	Enter the confirmed Chinese and English names.	HEALTH (Digital Health & Medical Image Analytics)
One-Sentence Definition	State directly what this direction studies.	Advanced AI and predictive analytics for diagnostic imaging and personalized health interventions.

Key Questions	List the core long-term questions.	1. Early prediction of Non-Communicable Diseases (NCDs) using complex sequence data. 2. High-precision cross-modal medical image segmentation. 3. Closed-loop control systems based on deep reinforcement learning (e.g., Artificial Pancreas).
Main Research Content	Enter the research objects, tasks, and related details.	1. Constructing disease intervention models using reinforcement learning. 2. Developing Dual-Branch CNN architectures for Alzheimer's diagnosis. 3. Multi-modal sentiment and emotion analysis for personalized health coaching.
Application Scenarios	List application environments that can be described clearly.	1. Automated medical image segmentation and clinical diagnostics. 2. Ageing-in-Place physiological and mental health monitoring. 3. Long-term dynamic risk prediction for diabetic patients.
Representative Projects	Enter confirmed P IDs.	P-07-01, P-07-02, P-07-06
Additional Notes	Important notes or additions.	Requires strict collaboration with clinical hospitals and public health agencies for data de-identification and validation.

R-03 EDGE-AI (Edge Computing & Vision)

Entry	Instructions / Suggested Length	Content
Name	Enter the confirmed Chinese and English names.	EDGE-AI (Edge Intelligence & On-Device Vision)

One-Sentence Definition	State directly what this direction studies.	Deploying high-performance computer vision and machine learning models on resource-constrained industrial and IoT devices.
Key Questions	List the core long-term questions.	1. Low-latency, real-time visual sensing independent of cloud infrastructure. 2. Real-time object and defect detection in complex manufacturing/traffic environments 3. Neural network compression and lightweighting for embedded hardware.
Main Research Content	Enter the research objects, tasks, and related details.	1. Developing lightweight Mamba-transformer hybrid architectures for real-world detection. 2. Zero-shot learning on gesture movement for interactive dashboard control.
Application Scenarios	List application environments that can be described clearly.	1. Smart retail checkout systems (e.g., AI weighing scales). 2. Autonomous drone mapping and image stitching for Agriculture 4.0. 3. Underwater sonar small target detection and traffic monitoring.
Representative Projects	Enter confirmed P IDs.	P-05, P-07-03, P-07-08, P-07-09
Additional Notes	Important notes or additions.	Supported by NVIDIA DLI Computer Vision certification, ensuring seamless integration with commercial AIoT hardware ecosystems.

R-04 AGENT (Multi-Agent Systems)

Entry	Instructions / Suggested Length	Content
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Name	Enter the confirmed Chinese and English names.	AGENT (Multi-Agent Systems & NLP)
One-Sentence Definition	State directly what this direction studies.	Large Language Model (LLM) driven autonomous system orchestration and multi-modal human-AI interaction.
Key Questions	List the core long-term questions.	1. Automating civic and smart tourism services via AI-RAG technology. 2. Enhancing digital interactions with emotionally intelligent conversational avatars. 3. Large-scale patent mining and text analysis powered by foundational models.
Main Research Content	Enter the research objects, tasks, and related details.	1. R&D of Retrieval-Augmented Generation (RAG) pipelines for public services. 2. Exploring genetic hybridization technology and emotional state transition models in human-robot interfaces. 3. Sustainable product innovation using TRIZ methodology combined with BERT and LLMs.
Application Scenarios	List application environments that can be described clearly.	1. National-level digital concierges and automated customer service systems. 2. Immersive multi-modal platforms for EdTech and psychological counseling. 3. Automated trend prediction and ideation for corporate R&D departments.
Representative Projects	Enter confirmed P IDs.	P-03, P-04, P-07-04, P-07-05, P-07-10

Additional Notes	Important notes or additions.	Strong focus on Federated Learning privacy mechanisms and commercial adaptability of LLM deployments.
R-05 Synergy of Research Pillars		
Entry	Instructions / Suggested Length	Content
Shared Problem	Describe the higher-level problem shared by all pillars.	All three pillars work together to bridge the gap between perceiving massive unstructured data in physical environments and executing real-time autonomous decisions in digital workflows.
Individual Boundaries	Describe the boundary of each of the three directions.	HEALTH: Strictly focuses on physiological data and clinical predictive modeling; does not handle general vision. EDGE-AI: Focuses on hardware-constrained, high-frequency visual processing; does not handle deep NLP logic. AGENT: Focuses on NLP, RAG knowledge bases, and conversational UI; does not handle raw sensor/hardware integration.
Connection Method	Explain how the three pillars connect and reuse one another.	EDGE-AI serves as the perception layer, capturing environmental or physical data at high speeds on constrained hardware. For biomedical scenarios, this data is passed to HEALTH algorithms for deep pathological or risk analysis. Finally, AGENT acts as the orchestration layer, synthesizing quantitative outputs from Edge and Health models into natural language advice or automated workflows (e.g., Digital Concierge or AI Coaching) for human end-users.

Relationship	Describe the	To be provided: A circular workflow diagram.
Visual	relationship diagram.	Edge-AI (Data Capture) at the base, Health (Domain Analysis) on the left, looping up to Agent (Human-AI Interaction) at the top.

R-06 Scenarios & Project Mapping

Entry	Instructions / Suggested Length	Content
HEALTH mapping	Use the format “scenario name: P ID”; do not repeat project descriptions.	Early Prediction of NCDs: P-07-01; Elderly Mental Health Monitoring: P-07-02; Personalized Fitness Guidance: P-07-06.
EDGE-AI mapping	Use the format “scenario name: P ID”; do not repeat project descriptions.	Smart Grocer Produce Recognition: P-05; Agriculture 4.0 Drone Mapping: P-07-03; Facial Recognition Access Control: P-07-08; Game Visual Guide: P-07-09.
AGENT mapping	Use the format “scenario name: P ID”; do not repeat project descriptions.	AI-RAG Digital Concierge: P-03; National Address System: P-04; Scam Monitoring: P-07-04; DingTalk Office Assistant: P-07-05; TikTok Lead Generation & Sales: P-07-10.
Additional Industry Applications	Do not force this into one of the three research pillars.	AI Art Image Generation Platform: P-07-07; Merace Metaverse Gaming Platform: P-07-11; Intelligent Insurance Services and Online Claims Platform: P-07-12; AI Social Commerce App: P-07-13; O2O Intelligent Matching and Activity-based Social Platform: P-07-14; RiceUp Local Food Delivery Platform: P-07-15; Mead Johnson Marketing Management System: P-07-16; Custom Warehouse Management System (WMS): P-07-17.

P-01 Projects Intro

Entry	Instructions / Suggested Length	Content
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Page Title	Enter the official title of the Projects page.	Key Projects & Tech Transfer
Homepage Summary	Summarize project value without elaboration.	Managing over RM 1.2M in research funding, our projects drive tangible B2B and enterprise value across smart manufacturing, digital health, and national civic infrastructure.
Inner-Page Introduction	Summarize what problems the lab addresses through projects.	Our projects represent the successful leap from academic breakthroughs to industrial applications. Backed by national grants and corporate co-funding (e.g., Advantech, HILTI), the HI Lab delivers highly adaptable AI systems tailored for predictive maintenance, edge deployment, and LLM orchestration. These solutions directly serve manufacturing enterprises, healthcare providers, and government policymakers. The directory also includes 13 company/team industry cases presented on the original KOALA SYNC TECH website as capability signals; these cases are not evidence of individually owned contributions, client relationships, or verified commercial metrics, and the statistical basis of all percentage metrics remains to be verified.

P-02 Capability Overview

Entry ID	Capability Category	Core Capabilities & Scenarios	Typical Deliverables & Related Projects
P-02-01	AI Solutions	Core capabilities: predictive analytics, medical AI, generative AI, RAG, agents, multimodal analysis, and	Typical deliverables: forecasting models, medical AI models, domain knowledge bases, RAG pipelines, agents, digital humans, and generative applications.

		marketing automation. Applicable scenarios: digital health, public services, manufacturing forecasting, and intelligent marketing.	Related projects: P-03, P-06, P-07-01, P-07-02, P-07-04, P-07-05, P-07-06, P-07-07, and P-07-10.
P-02-02	Edge Intelligence	Core capabilities: computer vision, model compression, on-device inference, AIoT, and hardware-software integration. Applicable scenarios: smart retail, agricultural mapping, access security, and game vision.	Typical deliverables: lightweight vision models, embedded inference solutions, real-time recognition systems, and IoT hardware integration. Related projects: P-05, P-07-03, P-07-08, and P-07-09.
P-02-03	Custom Digital Products	Core capabilities: Web, mobile apps, industry platforms, transaction workflows, recommendation, and user engagement. Applicable	Typical deliverables: custom Web and mobile applications, industry service platforms, transaction and content workflows, and recommendation and engagement modules. Related projects: P-07-11,

		scenarios: insurance, e-commerce, social networking, food delivery, and gaming platforms.	P-07-12, P-07-13, P-07-14, and P-07-15.
P-02-04	Digital Transformation	Core capabilities: enterprise systems, public digital infrastructure, process automation, data operations, and systems integration. Applicable scenarios: public governance, marketing management, warehousing, inventory, orders, and logistics.	Typical deliverables: digital infrastructure design, enterprise management systems, automated workflows, data analytics and reporting, and platform integration. Related projects: P-04, P-07-16, and P-07-17.

P-07 All Projects & Industry Application Cases

AI Solutions (9 projects)

Entry	Instructions / Suggested Length	Content (P-03)
Project Name	Enter the official project name.	Integrated Digital Concierge with Real-Time Avatar Based on AI-RAG

S — Situation	Describe the setting, current state, and constraints.	Current smart city and tourism services lack scalable, low-latency, and highly interactive touchpoints, making it difficult for citizens to retrieve civic information efficiently.
T — Task	Describe the objective and the team's responsibility.	Develop and deploy a national-level conversational AI avatar using Retrieval-Augmented Generation (RAG) to provide accurate public services and civic empowerment.
A — Action	Enter the concrete actions.	1. Built a domain-specific LLM knowledge base for civic policies. 2. Developed real-time 3D avatar rendering with multi-lingual support. 3. Optimized the RAG data pipeline for millisecond-level responsiveness.
R — Result	Enter the public result.	Successfully funded and actively deployed as a core civic empowerment tool (2026-2027).

Entry	Instructions / Suggested Length	Content (P-06)
Project Name	Enter the official project name.	Forecasting Energy Consumption for Predictive Maintenance
S — Situation	Describe the setting, current state, and constraints.	Heavy manufacturing relies on reactive maintenance, causing sudden equipment failures, massive energy waste, and unpredictable downtime costs.

T — Task	Describe the objective and the team's responsibility.	Build a data-driven deep learning model to accurately predict energy spikes and identify potential equipment failure points in advance.
A — Action	Enter the concrete actions.	1. Captured and cleansed high-frequency sensor data from the factory floor. 2. Applied adaptive dual distillation frameworks to optimize prediction accuracy. 3. Developed immersive Visual Analytics Dashboards for floor managers.
R — Result	Enter the public result.	Supported by the TS Leo Moggie Distinguished Chair Grant (RM 60,000), successfully optimizing maintenance schedules for roofing manufacturing plants.

Entry	Instructions / Suggested Length	Content (P-07-01)
Project Name	Enter the official project name.	Early Prediction of NCDs from Imbalanced Sequence Data
S — Situation	Describe the setting, current state, and constraints.	Non-Communicable Diseases (NCDs) have long incubation periods, and existing medical datasets are highly imbalanced, leading to low sensitivity in traditional early-warning models.
T — Task	Describe the objective and the team's responsibility.	Develop dynamic data augmentation algorithms suitable for highly imbalanced medical data to enhance early disease prediction capabilities.
A — Action	Enter the concrete actions.	1. Extracted clinical datasets containing sequence structures. 2. Designed dynamic data augmentation algorithms specific to sequence features.

		3. Validated early-warning accuracy via rigorous cross-validation.
R — Result	Enter the public result.	Funded by the Young Researcher Incentive Grant (RM 29,100), achieving breakthroughs published in Q1/Q4 SCIE journals.

Entry	Instructions / Suggested Length	Content (P-07-02)
Project Name	Enter the official project name.	Facial Expression Recognition for Elderly Mental Health Monitoring
S — Situation	Describe the setting, current state, and constraints.	The ageing-in-place population lacks objective monitoring tools for emotional and mental health, preventing timely psychological interventions.
T — Task	Describe the objective and the team's responsibility.	Develop a non-contact facial emotion recognition system integrated with an empathetic conversational agent for 24/7 mental state assessment.
A — Action	Enter the concrete actions.	1. Built a specialized facial feature dataset for elderly expressions. 2. Trained fuzzy multi-modal transformer models to analyze emotional transitions. 3. Output real-time psychological alerts to caregivers via dialogue systems.
R — Result	Enter the public result.	Successfully constructed emotional state transition models, with core findings published in <i>Expert Systems with Applications</i> .

Entry	Instructions / Suggested Length	Content (P-07-04)
Project Name	Enter the official project name.	Scam Surveillance Model Integrating Web Scraping and LLMs
S — Situation	Describe the setting, current state, and constraints.	Online scams evolve rapidly, and traditional keyword-based filters fail to identify deep-level misinformation and fraudulent intent in multi-modal formats.
T — Task	Describe the objective and the team's responsibility.	Build a fully automated scam surveillance and early-warning system by merging image forensics with Large Language Models.
A — Action	Enter the concrete actions.	1. Deployed web scrapers to continuously harvest multi-modal scam samples. 2. Fused ELA-ResNeXt image technology with BERT linguistic semantics. 3. Trained a joint classifier for detecting multi-modal fraudulent information.
R — Result	Enter the public result.	Backed by an Industrial Grant, significantly improving the detection rate of fraudulent information in complex online environments.

Entry	Instructions / Suggested Length	Content (P-07-05)
Project Name	Enter the official project name.	DingTalk AI Office Assistant
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, omissions and delays in daily tasks, schedules, and meetings can reduce team collaboration efficiency.

T — Task	Describe the objective and the team's responsibility.	Build an AI office assistant integrated with DingTalk to centrally manage tasks, schedules, and meeting reminders.
A — Action	Enter the concrete actions.	1. Integrated task and schedule management. 2. Configured reminders for meetings and action items. 3. Automated routine office tasks with AI and connected them to collaborative workflows.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, employee productivity increased by 30.33% and meeting attendance increased by 25.67%; the measurement methodology remains to be verified.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/solu_ban1.png Use: Project card or case-detail hero image Alt text: DingTalk AI Office Assistant showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.1; company/team showcase project, not evidence of individual sole authorship; no application source code, APIs, or acceptance materials were provided.
Entry	Instructions / Suggested Length	Content (P-07-06)
Project Name	Enter the official project name.	AI Fitness Assistant
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, differences in users' health conditions and exercise habits make generic training plans insufficient for personalized needs.

T — Task	Describe the objective and the team's responsibility.	Generate personalized fitness recommendations from health data, physical condition, and exercise habits, and help users set and track goals.
A — Action	Enter the concrete actions.	1. Consolidated user health information and exercise habits. 2. Built user profiles and recommended fitness plans. 3. Supported goal setting, execution, and feedback.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, user satisfaction increased by 40.13% and platform subscriptions grew by 20.68%; the measurement methodology remains to be verified.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/solu_ban2.png Use: Featured-project card or case-detail hero image Alt text: AI Fitness Assistant showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.1; company/team showcase project, not evidence of individual sole authorship; the data fields, recommendation algorithms, models, and privacy handling were not disclosed.
Entry	Instructions / Suggested Length	Content (P-07-07)
Project Name	Enter the official project name.	AI Art Image Generation Platform
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, personalized art creation has a high tool barrier, and users need a more direct way to generate content.

T — Task	Describe the objective and the team's responsibility.	Build an art image generation platform that supports text descriptions and style selection across multiple painting styles and themes.
A — Action	Enter the concrete actions.	1. Provided text-prompt and style-selection inputs. 2. Organized image-generation workflows across multiple styles and themes. 3. Supported the display of and interaction with user-generated content on the platform.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, user-generated content increased by 50.68% and platform engagement increased by 35.31%; the measurement methodology remains to be verified.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/solu_ban3.png Use: Project card or case-detail hero image Alt text: AI Art Image Generation Platform showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.1; company/team showcase project, not evidence of individual sole authorship; model selection, inference services, training methods, and content-safety mechanisms were not disclosed.
Entry	Instructions / Suggested Length	Content (P-07-10)
Project Name	Enter the official project name.	TikTok AI Lead Generation and Sales Automation System
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, identifying, following up, and converting potential leads in short-video comments is inefficient.

T — Task	Describe the objective and the team's responsibility.	Build a lead-generation and sales workflow covering video and comment collection, AI lead analysis, and automated follow-up.
A — Action	Enter the concrete actions.	1. Collected and downloaded video links and scraped comments. 2. Used AI to analyze comments and identify potential leads. 3. Automatically tracked leads and sent preset messages.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, customer conversion increased by approximately 30%; the measurement methodology, scraping authorization, privacy handling, and compliance with platform rules remain to be verified. TikTok is the target platform, not a verified client.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/solu_ban6.png Use: Project card or case-detail hero image Alt text: TikTok AI Lead Generation and Sales Automation System showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.1; company/team showcase project, not evidence of individual sole authorship; no business-system source code or independent acceptance materials were provided.

Edge Intelligence (4 projects)

Entry	Instructions / Suggested Length	Content (P-05)
Project Name	Enter the official project name.	Smart Grocer: Edge AI Produce Recognition System

S — Situation	Describe the setting, current state, and constraints.	Traditional supermarket checkouts rely heavily on cashiers manually identifying and entering codes for fresh produce, leading to bottlenecks and inventory errors.
T — Task	Describe the objective and the team's responsibility.	Integrate Edge AI vision models directly into retail weighing scales to achieve fully automated, real-time identification of fruits and vegetables.
A — Action	Enter the concrete actions.	1. Curated high-quality image datasets validated by agricultural taxonomists. 2. Trained and compressed lightweight vision classifiers for edge chips. 3. Integrated the software seamlessly with IoT scale hardware.
R — Result	Enter the public result.	Eliminated manual entry and drastically improved checkout speeds; key findings published in <i>IET Image Processing</i> .

Entry	Instructions / Suggested Length	Content (P-07-03)
Project Name	Enter the official project name.	Autonomous Drone Mapping with Wireless Charging for Agriculture 4.0
S — Situation	Describe the setting, current state, and constraints.	Large plantations rely on manual labor or short-battery drones for inspection, resulting in low data frequency and poor image stitching for precision agriculture.
T — Task	Describe the objective and the team's responsibility.	Deploy an autonomous drone network equipped with wireless charging, processing automated field image stitching and photogrammetry mapping at the edge.

A — Action	Enter the concrete actions.	1. Configured autonomous drone flight paths and wireless charging stations. 2. Applied computer vision algorithms for real-time agricultural image stitching. 3. Established a cloud analytics platform to generate precision farming indices.
R — Result	Enter the public result.	Secured RM 128,600 in internal funding, enabling fully automated aerial surveillance and high-precision mapping of large plantations.

Entry	Instructions / Suggested Length	Content (P-07-08)
Project Name	Enter the official project name.	AI Facial Recognition Access Control System
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, identity verification, security monitoring, and access control face efficiency and cost challenges.
T — Task	Describe the objective and the team's responsibility.	Use facial recognition for identity verification and access control in security monitoring and facility access scenarios.
A — Action	Enter the concrete actions.	1. Captured and matched facial identity information. 2. Connected recognition results to access control. 3. Linked security monitoring with facility access workflows.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, the system can improve security-management efficiency and reduce costs, but no quantitative metrics were provided; accuracy, false-match rate, liveness detection, response time, and hardware-integration evidence remain to be verified.

Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/solu_ban4.png Use: Featured-project card or case-detail hero image Alt text: AI Facial Recognition Access Control System showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.1; company/team showcase project, not evidence of individual sole authorship; no independent acceptance materials were provided.
Entry	Instructions / Suggested Length	Content (P-07-09)
Project Name	Enter the official project name.	GTorque Game Guide Assistant
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, players may struggle to identify key scenes, missions, or progression paths.
T — Task	Describe the objective and the team's responsibility.	Use image recognition and computer vision to understand game screens and provide players with guides and contextual tips.
A — Action	Enter the concrete actions.	1. Identified key scenes and missions in games. 2. Associated them with relevant guides and tips. 3. Delivered assistance as players progressed.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, the solution can improve player engagement, satisfaction, and product competitiveness, but no quantitative metrics were provided; supported games, recognition latency, and accuracy remain to be verified.
Project Image	Record only the asset path, use, alt text, and	../docs/assets/company-website/source/src/images/solu_ban5.png

	copyright status; do not embed the image.	Use: Project card or case-detail hero image Alt text: GTorque Game Guide Assistant showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.1; company/team showcase project, not evidence of individual sole authorship; the recognition model and business-system source code were not disclosed.

Custom Digital Products (5 projects)

Entry	Instructions / Suggested Length	Content (P-07-11)
Project Name	Enter the official project name.	Merace Metaverse Gaming Platform
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, the project addresses a gaming-platform and cross-game asset scenario involving blockchain, Binance Smart Chain, the metaverse, and DApps.
T — Task	Describe the objective and the team's responsibility.	Build a gaming platform that supports the purchase of platform currency, game characters, and items, as well as cross-game asset management and trading.
A — Action	Enter the concrete actions.	1. Organized platform-currency purchase and usage flows. 2. Supported trading of game characters and items. 3. Integrated multiple blockchain games and managed cross-game assets.
R — Result	Enter the public result; self-reported metrics	According to the original website materials, monthly active users increased by 200% and transaction volume increased by 34.48%; the

	must be marked for verification.	measurement methodology remains to be verified, and the repository provides no smart contracts, on-chain addresses, wallet integration, or security-audit evidence.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/what_right1.jpg Use: Project card or case-detail hero image Alt text: Merace Metaverse Gaming Platform showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.2; company/team showcase project, not evidence of individual sole authorship; no independent acceptance materials were provided.
Entry	Instructions / Suggested Length	Content (P-07-12)
Project Name	Enter the official project name.	Intelligent Insurance Services and Online Claims Platform
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, the experience of insurance-product discovery, online enrollment, customer consultation, and online claims needed improvement.
T — Task	Describe the objective and the team's responsibility.	Build a Web platform covering insurance product information, online enrollment, intelligent customer service, online consultation, and claim submission.
A — Action	Enter the concrete actions.	1. Integrated insurance-product and customer-service information. 2. Built online enrollment, consultation, and intelligent customer-service workflows. 3. Added online claim submission and status processing.

R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, online enrollment increased by 40.36%, customer satisfaction by 30.1%, customer-service response speed by 50%, issue-resolution rate by 45.43%, average daily visits by 60.37%, time on page by 20%, and successful online claim submissions by 35.64%; the measurement methodology remains to be verified.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/what_right2.jpg Use: Project card or case-detail hero image Alt text: Intelligent Insurance Services and Online Claims Platform showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.2; company/team showcase project, not evidence of individual sole authorship; the source-code title conflicts with the insurance-focused body copy, so the project is named according to the body content.
Entry	Instructions / Suggested Length	Content (P-07-13)
Project Name	Enter the official project name.	AI Social Commerce App
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, social commerce needs stronger user engagement, product discovery, and customer-conversion efficiency.
T — Task	Describe the objective and the team's responsibility.	Integrate new retail, social interaction, and product recommendation into a mobile shopping experience.

A — Action	Enter the concrete actions.	1. Supported product discovery and purchasing. 2. Enabled users to share shopping experiences and recommend products to friends. 3. Used interaction data for product recommendations and targeted marketing.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, platform engagement increased by 56.31% and sales increased by 27.2%; the measurement methodology and the AI model, data, and feature implementation remain to be verified.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/what_right3.jpg Use: Project card or case-detail hero image Alt text: AI Social Commerce App showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.2; company/team showcase project, not evidence of individual sole authorship; no business-system source code or independent acceptance materials were provided.
Entry	Instructions / Suggested Length	Content (P-07-14)
Project Name	Enter the official project name.	O2O Intelligent Matching and Activity-based Social Platform
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, users need a more efficient way to match with people who share their interests and participate in activities.
T — Task	Describe the objective and the team's responsibility.	Use intelligent matching algorithms and an activity-management system to improve matching, interaction, and event participation.

A — Action	Enter the concrete actions.	1. Established user-interest and matching mechanisms. 2. Provided flows for creating, managing, and joining activities. 3. Used interaction feedback to improve matching and activity experiences.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, successful matching increased by 45.4% and activity participation increased by 30.26%; the measurement methodology remains to be verified. The source expands O2O as Online-to-Online, which differs from the common interpretation.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/what_right4.jpg Use: Project card or case-detail hero image Alt text: O2O Intelligent Matching and Activity-based Social Platform showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.2; company/team showcase project, not evidence of individual sole authorship; no matching algorithm or independent acceptance materials were provided.
Entry	Instructions / Suggested Length	Content (P-07-15)
Project Name	Enter the official project name.	RiceUp Local Food Delivery Platform
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, users need a convenient way to discover nearby restaurants and place delivery orders.

T — Task	Describe the objective and the team's responsibility.	Build a local dining platform that aggregates nearby food-delivery services, personalized recommendations, and user reviews.
A — Action	Enter the concrete actions.	1. Aggregated nearby restaurants and delivery services. 2. Supported ordering, personalized recommendations, and reviews. 3. Used feedback to improve restaurant discovery and repeat ordering.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, order rate increased by 52.12%, satisfaction by 25.34%, active users by 35.46%, and new registrations by 40%; average weekly orders rose from two to three, and the positive-review rate reached 90%. The measurement methodology remains to be verified.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/what_right5.jpg Use: Project card or case-detail hero image Alt text: RiceUp Local Food Delivery Platform showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.2; company/team showcase project, not evidence of individual sole authorship; 90% is the positive-review rate, not a growth rate.

Digital Transformation (3 projects)

Entry	Instructions / Suggested Length	Content (P-04)
Project Name	Enter the official project name.	Technical and Operation Design for National Address Systems (NAS)

S — Situation	Describe the setting, current state, and constraints.	To accelerate the digital economy, Malaysia urgently needed a unified, intelligent addressing infrastructure to support logistics, resource allocation, and policymaking.
T — Task	Describe the objective and the team's responsibility.	Design the overall architecture, visual-based AI use cases, and governance framework for the National Address System.
A — Action	Enter the concrete actions.	1. Developed use cases for address data capture. 2. Formulated capacity building and training tools for the NAS system. 3. Established the technological and policy compliance framework (AI Enablers).
R — Result	Enter the public result.	Secured multiple industrial grants (RM 50k+ each) from Three-Opp (M) Sdn. Bhd., laying the foundation for Malaysia's digital transformation.

Entry	Instructions / Suggested Length	Content (P-07-16)
Project Name	Enter the official project name.	Mead Johnson Marketing Management System
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, audience segmentation, targeted marketing, user engagement, and marketing decision-making needed improvement.
T — Task	Describe the objective and the team's responsibility.	Build a marketing management system combining audience segmentation, targeted outreach, engagement points, data analytics, and reporting.
A — Action	Enter the concrete actions.	1. Performed personalized audience segmentation and targeting. 2. Configured

		engagement-point allocation. 3. Used integrated analytics and reporting to identify market demand and user preferences.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, the system can improve user engagement, brand loyalty, and sales, but no quantitative metrics were provided; the client relationship has not been independently verified through contracts or third-party materials.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/inf_banner.jpg Use: Project card or case-detail hero image Alt text: Mead Johnson Marketing Management System showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.3; project shown on a company/team website, not evidence of individual sole authorship; no independent acceptance materials were provided.
Entry	Instructions / Suggested Length	Content (P-07-17)
Project Name	Enter the official project name.	Custom Warehouse Management System (WMS)
S — Situation	Describe the setting, current state, and constraints.	According to the original website materials, manual records and legacy SaaS systems made warehouse operations inefficient and prone to human error.
T — Task	Describe the objective and the team's responsibility.	Use a custom WMS to integrate inventory, orders, logistics tracking, real-time monitoring, and automated operations.
A — Action	Enter the concrete actions.	1. Integrated inventory and order processing. 2. Connected logistics tracking and real-time data

		monitoring. 3. Used automation to reduce manual processes and errors.
R — Result	Enter the public result; self-reported metrics must be marked for verification.	According to the original website materials, process efficiency improved by 100%; the statistical definition remains to be verified and is not interpreted as halving processing time or doubling efficiency. The second carousel image also does not show an independent client name.
Project Image	Record only the asset path, use, alt text, and copyright status; do not embed the image.	../docs/assets/company-website/source/src/images/inf_right.jpg Use: Project card or case-detail hero image Alt text: Custom Warehouse Management System showcase image Copyright status: To be confirmed
Source / Evidence Metadata	State the source, attribution, and evidence boundary.	Source: docs/项目与优势.md §4.3; company/team showcase project, not evidence of individual sole authorship; no business-system source code or independent acceptance materials were provided.

P-08 Projects CTA

Fixed Item	Fixed Content
Title	Welcome Collaboration
Description	If you represent an enterprise seeking to deploy Edge AI, or an institution looking for joint R&D in Medical AI or Agent systems, please explore our collaboration options.
Button	View Collaboration Options
Target	Contact page

A-01 Intelligent Data Analytics

Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	Predictive Analytics & Knowledge Distillation
Method	Describe the key method and applicable conditions.	To combat the data scarcity and severe imbalance often found in industrial and medical datasets, we utilize adaptive dual distillation frameworks and dynamic data augmentation, transferring the complex reasoning of deep neural networks into highly robust, lightweight models.
Advantage	Describe the specific advantage and applicability boundaries.	Maintains exceptionally high predictive accuracy even in noisy or resource-constrained industrial environments. This capability is validated by official SAS Predictive Modeler certifications and successful B2B deployments.

A-02 Applied Agentic AI

Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	LLM & Multi-modal Emotional Orchestration
Method	Describe the key method and applicable conditions.	By deeply coupling Retrieval-Augmented Generation (RAG) with localized industry knowledge bases, and innovatively integrating Fuzzy Multi-modal Transformers with genetic hybridization, our AI not only generates text but performs emotional reasoning and state transitions.

Advantage	Describe the specific advantage and applicability boundaries.	Empowers AI systems with profound "context-awareness" and "empathy," entirely stripping away the mechanical feel of traditional bots. Perfect for high-touch environments like civic digital concierges and personalized health coaching.
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A-03 Edge AI		
Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	Hardware-Optimized Edge Vision Detection
Method	Describe the key method and applicable conditions.	Employing lightweight Mamba-transformer hybrids and generalized Gaussian distribution algorithms, we extreme-compress massive computer vision models (like YOLO), allowing them to run inference directly on embedded chips in scales, drones, or underwater sonar.
Advantage	Describe the specific advantage and applicability boundaries.	Achieves millisecond-level, real-time visual perception completely detached from cloud reliance, resolving network latency and privacy issues. Backed by NVIDIA DLI certification, ensuring seamless integration into mainstream AIoT hardware ecosystems.

A-04 Cross-Model Skill Compilation & Runtime Governance		
Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	Cross-Model Skill Compilation & Runtime Governance

Entry	Instructions / Suggested Length	Content
Method	Describe the key method and applicable conditions.	We treat Skills as compilable natural-language programs. Based on the capability profiles of models, Agent frameworks, and host environments, we perform ahead-of-time (AOT) adaptation, dependency binding, concurrent workflow orchestration, and runtime feedback optimization, while retaining version rollback capability.
Advantage	Describe the specific advantage and applicability boundaries.	Reduces repetitive rewrites, blind retries, and environment failures when migrating Skills across models and frameworks, making task success rates, latency, token consumption, and failure causes easier to track and optimize.

A-05 User-Controlled Layered AI Memory

Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	User-Controlled Layered AI Memory
Method	Describe the key method and applicable conditions.	We separate conversational working memory, structured long-term memory, and the proactive suggestion layer. Rule-based routing, explicit associations, source records, expiry reviews, and deletion mechanisms are combined to manage the lifecycle of memory storage, retrieval, and removal.

Entry	Instructions / Suggested Length	Content
Advantage	Describe the specific advantage and applicability boundaries.	Reduces the need for users to repeat information while keeping sources traceable, content editable, permissions controllable, and expired memories removable. If long-term memory is unavailable, the system can still fall back to a standard single-session conversation.

A-06 Utility–Diversity Training Data Selection

Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	Utility–Diversity Training Data Selection
Method	Describe the key method and applicable conditions.	We reuse logits produced by the supervised fine-tuning forward pass and apply the nuclear norm to estimate sample utility and within-sample diversity. Low-dimensional representations and a historical buffer then identify redundancy across samples, enabling online selection of data that is more valuable for parameter updates.
Advantage	Describe the specific advantage and applicability boundaries.	Prioritizes backpropagation for high-value, non-redundant samples, increasing the density of effective updates within a fixed training budget and reducing the compute and time wasted on repeatedly training with homogeneous data.

A-07 On-Device Agents & Hardware Co-Design

Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	On-Device Agents & Hardware Co-Design

Entry	Instructions / Suggested Length	Content
Method	Describe the key method and applicable conditions.	On resource-constrained devices such as the ESP32-S3, an event-driven Agent Loop connects sensors, local structured memory, and actuators, while MCP Client / Server interfaces provide access to authorized services and hardware endpoints.
Advantage	Describe the specific advantage and applicability boundaries.	Allows low-risk events to be handled closer to the device, reduces dependence on cloud-only connections, and lowers the cost of building one-off interfaces among sensors, actuators, and agent platforms.

A-08 Verifiable Feedback-Driven Continual Learning

Entry	Instructions / Suggested Length	Content
Name	Enter the technology advantage name.	Verifiable Feedback-Driven Continual Learning
Method	Describe the key method and applicable conditions.	We anchor training direction with verifiable rewards, combine RLSD fine-grained self-distillation, NPO near-future policy guidance, and CoPD multi-expertise collaborative training, and record validators, rewards, model versions, and per-dimension evaluation results.
Advantage	Describe the specific advantage and applicability boundaries.	Provides denser learning signals that the current model can absorb more readily, reduces ineffective updates and conflicts when merging expertise, and makes continual model iteration more interpretable, measurable, and reversible.

N-01 Partners Intro

Entry	Instructions / Suggested Length	Content
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Shared Homepage and Partners-Page Intro	Briefly state which organizations are included.	The HI Lab maintains close, active collaborations with global tech leaders and top-tier academic institutions to drive the R&D and deployment of Edge Computing and Medical AI.
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N-02 Partners Logo Wall

Entry ID	Organization Name	Logo File or Location	Logo Public-Use Permission
N-02-01	Advantech	TBD (Advantech Logo)	Yes
N-02-02	HILTI Asia IT Services	TBD (HILTI Logo)	No
N-02-03	Tokio Marine & Dynafront	TBD (Tokio Marine Logo)	No
N-02-04	Xiamen University Malaysia	TBD (Xiamen Univ Logo)	No
N-02-05	Leeds Beckett University	TBD (Leeds Logo)	No
N-02-06	Three-Opp (M) Sdn. Bhd.	TBD (Three-Opp Logo)	No
N-02-07	Universiti Tenaga Nasional (UNITEN)	TBD (UNITEN Logo)	Yes
N-02-08	Asia Roofing Industry	TBD (Asia Roofing Logo)	No
N-02-09	Mobiva	TBD (Mobiva)	Yes

T-01 Team Intro

Entry	Instructions / Suggested Length	Content
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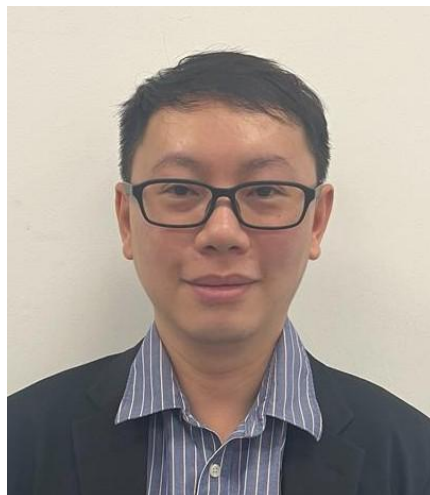
Page Title	Enter the official team-page title.	Research Team & Academic Network
Homepage Summary	Summarize the team composition.	Led by Dr. Chaw Jun Kit, our dynamic team of researchers bridges the gap between complex AI algorithms and commercial adaptability.
Inner-Page Introduction	Summarize the team's role.	The HI Lab brings together brilliant minds specializing in Computer Vision, Applied Machine Learning, and AI System Design. Our members focus intensively on Digital Health, On-device Intelligence, and LLM Orchestration. Through rigorous, hands-on R&D training, our alumni not only publish in top-tier international journals but also secure key positions in leading tech firms and prominent academic institutions.

T-02 PI Profile		
Entry	Instructions / Suggested Length	Content
Name	Chinese and English names and display order.	Dr. Chaw Jun Kit
Title / Role	Official title and role.	Principal Investigator (PI) / Senior Lecturer & Research Fellow
Affiliation	Official institutional affiliation.	Institute of Visual Informatics (IVI), Universiti Kebangsaan Malaysia (UKM)

Research Interests	Research-interest keywords.	Machine Learning, Intelligent Data Analytics, Edge AI, Computer Vision.
Short Bio	Educational or professional background and responsibilities.	Dr. Chaw holds a PhD in Electrical Engineering from UTM. He is a SAS-certified Predictive Modeler and an NVIDIA Deep Learning Institute (DLI) certified Computer Vision expert. His work uniquely bridges academia and industry, having managed over RM 1.2M in grants and served as an active tech collaborator for Advantech and HILTI, specializing in manufacturing optimization and digital health solutions.
Research Areas	Cite confirmed R IDs.	R-02 HEALTH, R-03 EDGE-AI, R-04 AGENT
Representative Projects / Outputs	Cite representative project or output IDs.	P-03 (Digital Concierge, Project Leader); P-05 (Smart Grocer); Published 50+ SCIE (Q1/Q2) & Scopus papers.
Personal Link	Personal or academic homepage.	UKMsarjana Academic Profile

Photo

Photo description.



Additional
Notes

Additional information.

Professional Member of the Board of Engineers
Malaysia (77508A) and an HRDCorp Accredited
Trainer.

Entry

**Instructions /
Suggested Length**

Content

Name

Chinese and English
names and display
order.

Dr. Jeff Wang

Title / Role

Official title and role.

Chief Technology Officer (CTO)

Affiliation

Official institutional
affiliation.

Research
Interests

Research-interest
keywords.

Short Bio

Educational or
professional
background and
responsibilities.

Research Areas	Cite confirmed R IDs.
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Representative Projects / Outputs	Cite representative project or output IDs.
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Personal Link	Personal or academic homepage.
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Photo	Photo description.
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Additional Notes	Additional information.
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Entry	Instructions / Suggested Length	Content
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Name	Chinese and English names and display order.	Wendy Leong Pooi Yan
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Title / Role	Official title and role.	CEO of Mobiva
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Affiliation	Official institutional affiliation.	Mobiva
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Research Interests	Research-interest keywords.	AI in Crisis Management, Pandemic Prediction, Enterprise Mobility, Cloud-based Decision Support.
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Short Bio	Educational or professional	Wendy Leong is the Chief Executive Officer of Mobiva, an industry-leading enterprise mobility
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	background and responsibilities.	and emergency management platform. Concurrently pursuing her PhD at IVI-Edge Lab, she uniquely bridges commercial tech leadership with advanced AI research. Her academic focus explores AI synergy in pandemic prediction and real-time, inclusive AI for disaster response, translating academic frameworks into scalable, enterprise-grade B2B solutions.
Research Areas	Cite confirmed R IDs.	R-02 HEALTH, R-04 AGENT
Representative Projects / Outputs	Cite representative project or output IDs.	Quantum Leaps in Pandemic Prediction (Book Chapter, 2025); Enterprise integration via Mobiva.
Personal Link	Personal or academic homepage.	LinkedIn Profile Mobiva Official Website
Photo	Photo description.	[Insert Professional Headshot of Wendy Leong]
Additional Notes	Additional information.	Brings extensive C-level B2B industry experience, driving the lab's commercial tech transfer and real-world disaster management implementations.

T-03 Team Composition			
Type	Count	Publication Status	Additional Notes
PhD Graduates	3	Public	2 completed as Main Supervisor
PhD Candidates	7	Public	Focusing on Deep Learning & Multimodal Analysis

MSc Graduates	3	Public	3 completed as Main Supervisor
MSc Candidates	2	Public	Focusing on Applied Algorithms

T-04 Core Members

Entry Content (T-04-01)

Name	Zhao Yanfeng
Status	PhD Graduate / Core Member
One-Sentence Bio	Focuses on safe-enhanced fully closed-loop artificial pancreas controllers based on Deep Reinforcement Learning.

Entry Content (T-04-02)

Name	Cheng Xiang
Status	PhD Graduate / Core Member
One-Sentence Bio	Specializes in adaptive dual distillation frameworks for efficient remaining useful life and predictive maintenance.

Entry Content (T-04-03)

Name	Liu Jianbang
Status	PhD Graduate / Core Member
One-Sentence Bio	Researches multimodal conversational emotion reasoning and immersive technology behavioral analysis.

Entry Content (T-04-04)

Name	Wang Hongqing
Status	PhD Candidate
One-Sentence Bio	Developer of Hybrid-YOLO architectures, focusing on complex traffic detection and medical image segmentation.

Entry **Content (T-04-05)**

Name Muhammad Aiman Md Zuki

Status PhD Candidate

One-Sentence Researches multi-modal sentiment and emotion analysis frameworks for
Bio personalized, persuasive health coaching.

Entry **Content (T-04-06)**

Name Leong Pooi Yan

Status PhD Candidate

One-Sentence Explores AI synergy in the prediction of future pandemics and automated
Bio crisis management tools.

Entry **Content (T-04-07)**

Name Zheng Kun

Status MSc Graduate

One-Sentence Primary researcher on generalized Gaussian distribution improvements and
Bio underwater sonar small target detection.

Entry **Content (T-04-08)**

Name Ooi Tze Yaang

Status MSc Graduate

One-Sentence Developed dynamic data augmentation methodologies with sequence
Bio structures for the early prediction of NCDs.

Entry **Content (T-04-09)**

Name	Teo Shi Han
Status	MSc Graduate
One-Sentence Bio	Developed Edge AI applications for real-time fresh produce identification in retail weighing systems.

C-01 Contact Intro

Entry	Instructions / Suggested Length	Content
Page Title	Enter the official title of the Contact page.	Partner With Us
Homepage Summary	Summarize suitable reasons to contact the lab.	We actively seek enterprise partnerships for Edge AI R&D, cross-disciplinary medical data collaborations, and highly motivated postgraduate (Master/PhD) candidates.
Inner-Page Introduction	Describe which collaboration inquiries are welcome.	The HI Lab balances rigorous academic research with strong commercial viability. Whether you are a manufacturing enterprise seeking predictive maintenance solutions, a clinical institution looking to leverage AI for diagnostics, or a prospective scholar passionate about Edge Computing and LLMs, we look forward to exploring collaborative opportunities with you.

C-02 Target & Collab Types

Entry ID	Collaboration Target	Collaboration Direction & Related IDs	Scope & Notes
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C-03-01	Industrial Tech Transfer (Joint R&D)	Enterprises facing specific challenges in automation, visual inspection, or energy efficiency.	Investment: Enterprise provides funding (e.g., Industrial Grants) and raw data; Lab delivers algorithms. Process: Needs Assessment -> NDA -> Tech Validation -> Project Kick-off. Homepage Display: Yes (Sort: 1) Note: PI directly assesses project feasibility.
C-03-02	Joint Academic Research & Grants	Universities or Institutes aiming for Q1 publications and national/international grants.	Investment: Sharing of core data, research personnel, and computing clusters. Process: Academic Alignment -> Proposal Drafting -> Joint Submission. Homepage Display: Yes (Sort: 2) Note: Cross-border and cross-disciplinary collaborations are highly welcomed.
C-03-03	Postgraduate Talent Recruitment	Exceptional students aiming for advanced AI degrees, top-tier publications, or careers in big tech.	Investment: Full-time commitment to research; Lab provides supervision, hardware, and publishing support. Process: CV/Proposal Submission -> Interview -> Official Faculty Admission. Homepage Display: Yes (Sort: 3)

Note: Comprehensive training from theory to engineering deployment provided.

C-04 Student Applications

Entry	Instructions / Suggested Length	Content
Recruitment Target	Enter the student type being recruited.	Highly self-motivated applicants for PhD and Master by Research with a background in CS, EE, or Data Science.
Direction	Enter the open research direction.	Edge AI Vision Deployment, LLM-Agent Orchestration, and Medical Image Analytics.
Application Method	Enter the required application materials.	Please send a detailed CV, academic transcripts, and a 1-page preliminary research proposal to chawjk@ukm.edu.my . Subject Line: [PhD/MSc Application - Your Name].
Current Status	Enter the recruitment status.	Long-term Open Recruitment (Official intakes follow the UKM academic calendar).

C-05 Contact Info

Entry	Instructions / Suggested Length	Content
Contact Person	Contact person's name.	Dr. Chaw Jun Kit (Principal Investigator)
Email	Official contact email.	chawjk@ukm.edu.my

Location	Approved public address.	Institute of Visual Informatics (IVI), Universiti Kebangsaan Malaysia (UKM), 43600 Bangi, Selangor, Malaysia.
Social / Booking	Social-media or academic homepage.	Visit UKMsarjana Profile

C-06 Communication Guidelines

Entry	Instructions / Suggested Length	Content
What to Provide When Contacting	What to provide in the initial communication.	Enterprise Inquiries: Briefly describe your business context, the specific technical pain point, data volume, and hardware constraints. Academic/Student Inquiries: Provide comprehensive educational background, past project outcomes, and a list of your core technical stack.
Matters We Respond To	Specify the inquiries we can handle.	We Respond To: Industrial grant alignments, joint academic research proposals, R&D consulting, and postgraduate admissions. We Do Not Respond To: Vague commercial proofs-of-concept without data backing, or general non-academic/non-industrial administrative queries.
Expected Response	Expected response time.	We conduct initial feasibility screenings for all collaboration and admission requests. Highly aligned inquiries will typically receive a response within 3-5 business days. If you do not hear back within two weeks, it indicates no current capacity for the proposed collaboration.